DepthWizard: Building, Tree and Ground Heights from One Satellite Image, with a Mamba-Based Height Model, for Disaster Response in India

HS202: Human Geography · Project Report

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Code and demo: https://github.com/hima1323/terravista-depthwizard

Abstract

When a flood or landslide strikes, rescue teams need to know how tall buildings are, how high the ground lies and where water will reach. Most of India has no recent 3D map that can answer this. DepthWizard estimates the height of every building, tree and patch of ground from one ordinary satellite photograph, ties those heights to sea level, and says how accurate they are. Our main new idea is a height model built on Mamba, a recent kind of neural network that reads an image in one sweep. It is nine times smaller and 5.2 times faster than the published model it learns from, although its first version is less accurate. Around the model we add a ground-level correction that uses NASA's satellite lasers, an error range for every building, and eighteen tools for disaster teams. With the published model, the average height error is 3.90 m on 150 aerial test tiles, where a flat-ground guess is off by 5.65 m, and 4.85 m at eight Indian sites. On the 2023 Teesta flood in Sikkim, free satellite images map 1.28 km^2 of damaged valley floor.

Keywords: height from a single image; Mamba; satellite lasers; flood mapping; disaster management; India.

3. Definition of the Problem

3.1 Problem statement

Estimate a reliable height, in metres, for every point of a single satellite photograph, and show how far that height can be trusted.

3.2 Origin/identification of the problem

The problem comes from how disasters are handled in India. In a flood, height decides the outcome: whether water enters a house, which roofs can shelter people, which slopes may slide. Yet the height data that planners need is rarely there. Laser surveys from aircraft are accurate but expensive, and

few Indian districts have one. Pairs of satellite images taken from two angles also give height, but a pair seldom exists for the place and date of an emergency. Free worldwide ground maps record one value every 30 m, which is too coarse to see a building. One satellite photograph, on the other hand, exists for almost every place and date. Recovering heights from that photograph would fill the gap almost everywhere.

In India the gap is wider. Floods, cloudbursts and landslides strike somewhere in the country every monsoon, often in hill towns and river-bank settlements that have grown quickly. The Teesta flood of October 2023 in Sikkim, which we study in Section 11.5, is one example. For that valley we could find no public height map detailed enough to show single buildings, and the free 30 m map was the best available.

3.3 Methodology adopted to identify the problem

We did not run surveys, key-informant interviews or focus-group discussions. We identified the problem in three ways. We read the published work on height from single images, on free ground maps and on satellite lasers. We discussed within the team what a disaster manager would need from a height map. And before building anything, we measured how far the free ground map sits from laser-measured ground at Indian sites. In towns it was 5.28 m off on average, which told us where to work first.

3.4 Detailed description of the identified problem

Height from one image is hard to use for three reasons. First, the image is ambiguous: seen from above, a tall building and a short one can look alike, so models guess from shadows and roof size and tend to make tall objects too short. Second, flood work needs height above sea level, while a model gives height above the ground; in towns and forests the free ground map includes roofs and tree tops, so its ground sits too high. Third, most tools give a height map without any error figure, and a relief officer cannot tell whether a 2 m flood depth is good to half a metre or to five. A useful tool must also work across India's very different landscapes: dense cities, scattered villages, hill towns and forests. A method tuned for flat cities can fail on a hillside, where the slope itself can be mistaken for height.

3.5 Current developments in the domain

Neural networks can now estimate depth from one photograph [5]. RS3DAda [3] adapted this to satellite images by training on computer-generated cities, and it is the model we start from. Depth Anything V2 [6] is a strong general model trained on ordinary photographs. Mamba [1] and MambaVision [2] are newer network designs that cost less to run. For the ground, Copernicus GLO-30 [8] is a free worldwide map and FABDEM [9] is a version with buildings and forest removed. NASA's ICESat-2 [10, 11] and GEDI [12] satellites measure heights with lasers along narrow tracks, and Song and others [7] recently showed that such laser points can correct single-image heights. Meta has published a worldwide map of tree heights [13]. India has its own Cartosat images and CartoDEM ground map from NRSC, but both need a login, so we have not tested them yet.

3.6 Need and significance of resolving the problem

With a height map that can be trusted, a district office could see before the monsoon which streets a given water rise would reach, route rescue teams away from deep water, choose helipad and relief-camp sites, and count affected buildings afterwards. Today these jobs rely on the coarse 30 m map or are not done. The need is greatest in the places with the least data: small towns, hill settlements and river banks, where a survey flight is unlikely to be paid for but a satellite photograph already exists. The error figure matters too. A planner who knows a map is good to 3 m in towns and poor under forest can still use it with care; with no error figure, people either trust the map completely or ignore it. Table 1 shows what each existing source lacks.

Table 1: Ways of getting heights for disaster work.

Approach	Needs	Detail	Sea level	Error stated
Aircraft laser survey	A survey flight	Very fine	Yes	Yes
Two-angle satellite pair	Two matched images	Fine	Yes	Partly
Free ground maps [8, 9]	Nothing	30 m	Yes	No
RS3DAda [3]	One image	Pixel	No	No
DepthWizard (ours)	One image and free data	Pixel	Yes	Yes

4. Aims/Objectives/Goals

Our aim is one offline tool that turns a single satellite photograph into a height map that disaster teams can use and check. Each objective in Table 2 closes one gap from Section 3 and has a test.

Table 2: Objectives and how each is measured.

#	Objective	How success is measured
O1	A height for every point of one image	Smaller error than a flat-ground guess on unseen data
O2	Heights above sea level	Closer to laser-measured ground than the free ground map
O3	An error range on each building	The range holds 90% of the time in unseen cities
O4	Checking inside the app	Error figures for each scene and type of land
O5	Disaster tools that are easy to find	Every tool opens from one search box
O6	Works offline on a laptop	Measured time for one full scene
O7	A cheaper height model (Mamba)	Speed, size and accuracy on the same tests as RS3DAda

5. Tools and Techniques (Interventions)

5.1 Height model

RS3DAda [3] gives, for every pixel, a height above the ground and a land-cover class such as building, tree or road. We cut large scenes into overlapping tiles and read each tile four times, flipped, then average the answers. How much the four answers disagree, u, tells us where the model is unsure.

We chose RS3DAda for three reasons. It was trained for satellite images, so it predicts height above the ground and not distance from a camera. It predicts land cover in the same pass, which the ground step below needs in order to find roofs and trees. And its trained weights are public, so anyone can repeat our numbers. We did not retrain it: every result in this report uses the released weights.

5.2 From height above ground to height above sea level

The model gives the height h of each point x above the ground under it. Adding the height of the ground gives the height above sea level:

$$\mathrm{DSM}(x) \;=\; \mathrm{DTM}(x) \;+\; h(x). \tag{1}$$

The ground map starts as GLO-30 [8], which in towns sits on rooftops. In each 30 m square we subtract the average building and tree height the model found there. We then shift the whole map

up or down to match ICESat-2 laser points that reached the ground. Two safety rules came from failures we saw. We refuse to shift when only one laser track crosses the scene, and we throw away "ground" laser points that sit on a roof. A user with a few surveyed points can correct the map further. If K points of known height z_k are available at positions (x_k, y_k), we find the level, scale and tilt that best fit the map's values $\hat{z}_k$ to them:

$$\min_{a,b,c,d} \sum_{k=1}^{K} \left(z_k - a - b\,\hat{z}_k - c\,x_k - d\,y_k \right)^2 . \tag{2}$$

One point sets only the level a. Two or three also set the scale b, and four or more also set the tilt c, d. Each fit reports its error on a point that was left out, so the user can see how much to trust it. This also lets a plain photograph without location data reach heights above sea level.

5.3 Tree heights

The model sees trees at only 18 to 60% of their true height. Trees can take their height from Meta's tree-height map [13], or be scaled up by the typical ratio between GEDI laser heights [12] and the model's heights at the same spots.

5.4 An error range for every building

We turn the disagreement u into a range with a method called split conformal prediction [17]. On buildings with known heights we compute the errors e_i and set

$$r_i = \frac{|e_i|}{u_i + u_0}, \qquad C(x) = \left[\hat{h}(x) \pm \hat{q}\,\left(u(x) + u_0 \right) \right], \tag{3}$$

where $\hat{q}$ is the value that 90% of the scores r_i fall below and $u_0 = 0.25\,\mathrm{m}$. The range $C(x)$ should then contain the true height nine times in ten, and we test that promise in cities that were not used to set it.

5.5 Disaster models

For a simple flood, every place below a chosen water level that connects to the river is marked wet. For water that moves over time we use the flow model of Bates and others [14]. It moves water between neighboring squares of the map according to the slope of the water surface, the depth of the water and the roughness of the ground. We added limits so that no square gives away more water than it holds. A real flood can be loaded as a ready-made map, or taken from a radar satellite image that the app splits into water and land [15]. To compare a predicted flood with a real one we use the critical success index,

$$\mathrm{CSI} \;=\; \frac{TP}{TP + FP + FN}, \tag{4}$$

where TP counts places both maps call wet, and FP and FN count places only one of them calls wet. A score of 1 is a perfect match. Evacuation routes are the cheapest paths [18] that avoid steep slopes, deep water and buildings, and landslide risk follows the Indian Standard IS 14496 [16]. The number of people at risk is estimated from each affected building's floor area, its number of floors, the share of it that is lived in, and the floor space one person needs. The app labels all of these as scenarios, not forecasts.

5.6 The application

DepthWizard runs on an ordinary laptop and works without internet once a scene has been opened. It has a scene library and six working pages that follow a disaster from start to finish, shown in

Figure 2. Table 3 lists its eighteen tools, which switch on in place, and any of them can be found by typing its name in a search box. The 3D view can be circled, flown through freely or walked at street level, and a flight along a drawn path can be recorded as a video for briefings. Results can be saved as standard map and 3D files that open in QGIS, ArcGIS or Bhuvan.

Table 3: The eighteen tools and the question each answers.

Tool	Question it answers
Explore	
Line of sight	What can be seen from this mast or tower?
Volume and area	How much debris is in this pile?
Drone flight plan	Which route keeps a drone above every roof?
Rooftop solar	How much power could each roof produce?
Map overlay	Where are the streets on the 3D view?
Mosaic	How do neighbouring scenes join up?
Validate and calibrate	
Heights from shadows	Do shadow lengths confirm the heights?
Tree and tall-object fix	Can weak heights be corrected from other sources?
Error bars	How far could each height be off?
Terrain source	Can I use my own ground map?
GPS and field points	Can points measured on the ground correct the map?
Prepare, Respond, Relief, Assess	
Landslide risk	Which slopes may fail in heavy rain?
People at risk	How many people live in the affected buildings?
Observed flood	How well does the predicted flood match a real one?
Flowing water	Where does the water go, hour by hour?
Evacuation plan	Which camp should each family go to?
Damage before/after	Which buildings collapsed?
Situation report	What does a decision maker need on one page?

6. Detailed Work Plan / Technological Interventions

Figure 1 shows the seven steps a scene goes through. An ordinary PNG or JPEG photograph gives heights above the ground. A GeoTIFF, which carries its location, continues to heights above sea level. Table 4 gives the order in which we built the parts, and Figures 2 and 3 show the result.

Most of the effort went into the fifth step. A height above the ground is of little use for floods until the ground itself is right, and the free ground map is wrong in exactly the places where people live. We also wrote automated tests that click through every tool on test scenes whose true answers are known: 38 checks in the main interface and 36 in the older viewer. All of them pass.

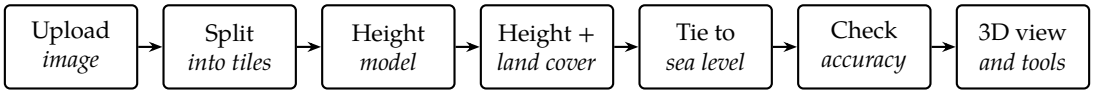


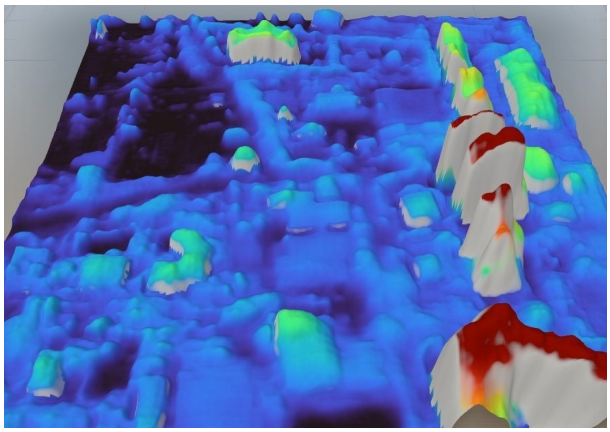
Figure 1: From one photograph to a checked 3D map.

Table 4: Work plan as carried out.

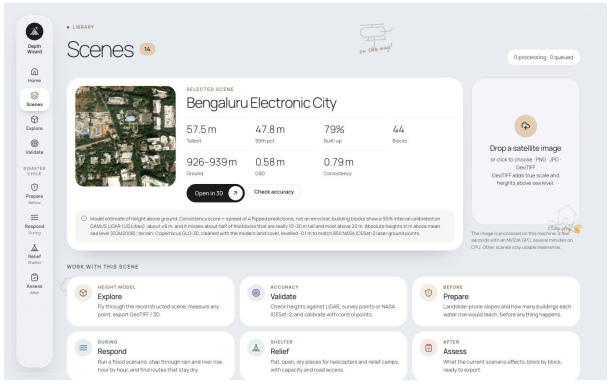
Period	Work
Before 15 October	Study of the problem, choice of model, first height maps
15 October	Checking inside the app, correction from surveyed points, installer, first disaster tools
18 October	Test method with confidence ranges, register of every number, error range per building
22 to 25 October	New interface, roof-point rule for lasers, training code for the Mamba model
28 October to 5 November	Eighteen tools and the tool finder, tree-height correction, Teesta case study, Mamba training and model comparison



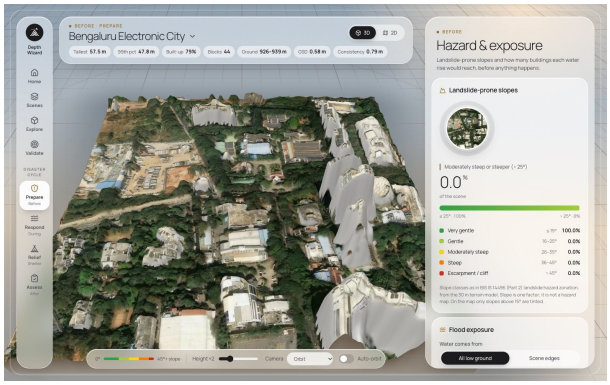
(a) 3D model with the photograph on it



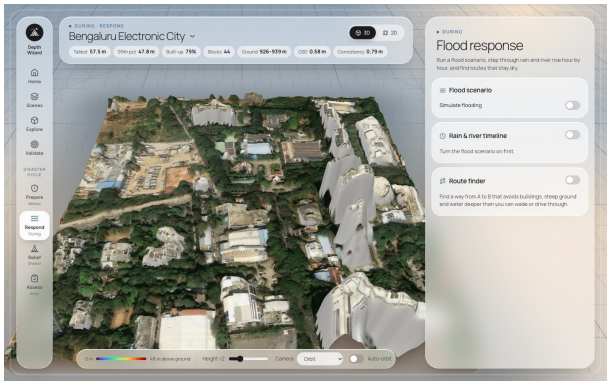
(b) Colored by height: 0 m blue, 48 m red



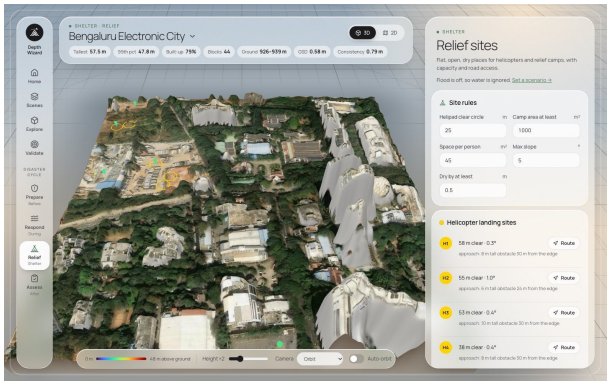
(c) Scene library and upload



(d) Prepare: risky slopes and water rise

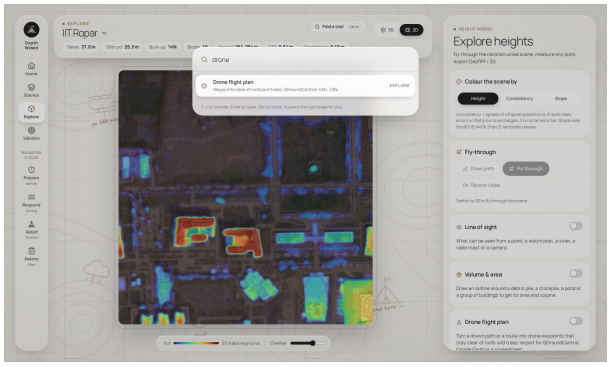


(e) Respond: flood scenario and routes

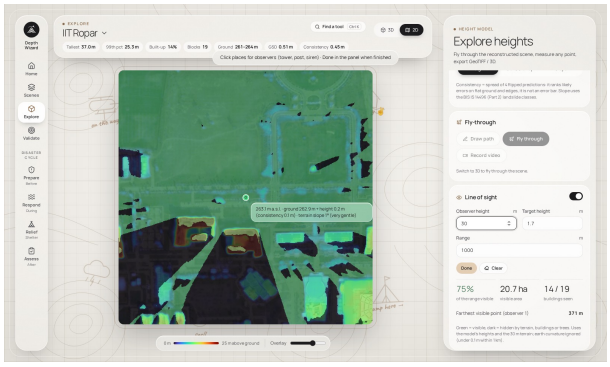


(f) Relief: helipad and camp sites

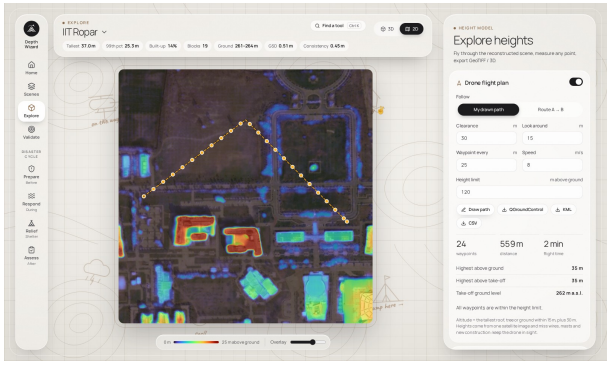
Figure 2: DepthWizard on a real scene, Bengaluru Electronic City, built from one satellite image. Heights in (a) and (b) are drawn at twice their true scale. Imagery: Esri World Imagery, for demonstration only.



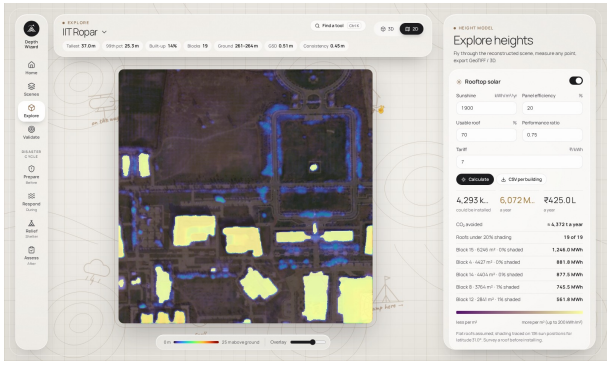
(a) Tool finder: typing “drone”



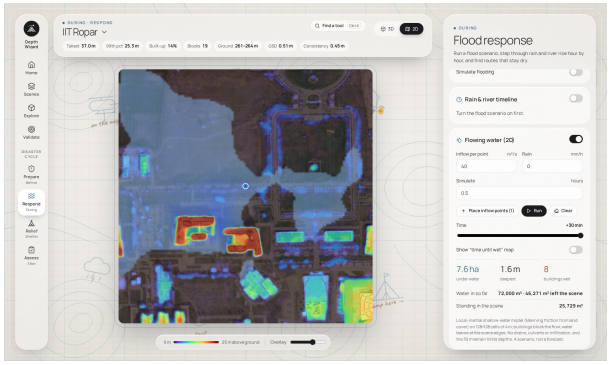
(b) Line of sight from a 30 m mast



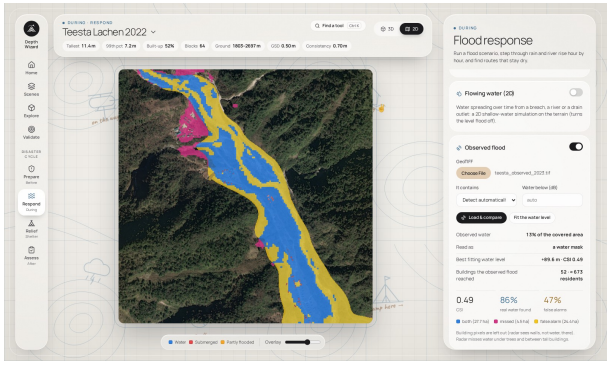
(c) Drone flight plan above every roof



(d) Rooftop solar with shading



(e) Flowing water from an inflow point



(f) Predicted against real flood, Teesta 2023

Figure 3: Tools running in the app. Panels (a) to (e): the IIT Ropar campus. Panel (f): the Teesta valley with the 2023 damage map.

7. Novelty / Innovation of the Proposed Interventions

7.1 A Mamba-based height model

RS3DAda reads images with a Transformer, a network that cuts the image into patches and compares every patch with every other patch. The work grows with the square of the image area, so large satellite scenes are slow and need a strong computer. Mamba [1] reads the patches one after another and keeps a short running memory, so its work grows only in step with the area.

We built a height model on MambaVision [2], a version of Mamba made for images, and joined it to RS3DAda’s two output parts, one for height and one for land cover. Figure 4 shows the layout. It gives the same kind of output as RS3DAda, so the rest of DepthWizard runs unchanged. A smaller and faster model matters for the users we have in mind. It can cover a whole district in the time the larger model needs for a few towns, and it needs far less graphics memory.

Training a height model from nothing needs a very large labeled dataset. We used knowledge

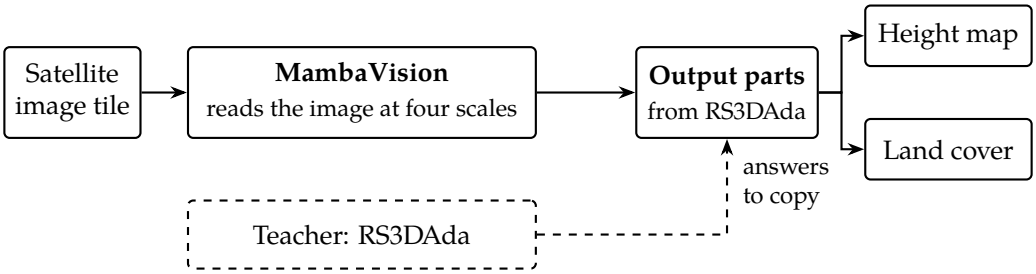


Figure 4: The Mamba-based height model and how it learns from RS3DAda.

distillation [4] instead: the Mamba model, called the student and written S, looks at unlabeled images and learns to give the same answers as RS3DAda, the teacher, written T. Its training goal is

$$\mathcal{L} \;=\; \ell\big(\hat{h}_{\mathrm{S}}, \hat{h}_{\mathrm{T}}\big) \;+\; \lambda\,\tau^{2}\,\mathrm{KL}\big(\sigma(z_{\mathrm{T}}/\tau)\,\big\|\,\sigma(z_{\mathrm{S}}/\tau)\big) \;+\; \gamma\,\ell\big(\hat{h}_{\mathrm{S}}, h_{\mathrm{true}}\big). \tag{5}$$

The first part pulls the student's heights towards the teacher's. The second does the same for land cover, with $\lambda = 0.5$ and $\tau = 2$. The third is optional and uses true heights from computer-generated cities, so that the student can do better than its teacher. We trained for 20,000 steps on tiles from the SynRS3D dataset. Section 11 reports how it did.

7.2 Other new parts

Most single-image tools stop at height above ground. We go on to height above sea level by cleaning the free ground map with the model's own output and levelling it with NASA lasers. Two safety rules in that step came from mistakes we saw: in Silent Valley one laser track along a forested valley floor would have moved the whole scene by 46.59 m, and in Chandigarh all 35 "ground" laser points we held back for checking were on roofs. The app now refuses both cases. Each building also gets an error range whose reliability we measured in cities the method never saw. Finally, every figure in this report is linked in a register to the file that produced it, with its confidence range and a simple guess to compare against.

8. Approaches that can be taken to Implement Intervention Plans

DepthWizard can reach users in five ways.

1. As an offline app for district and state offices. It installs with one click. One full scene took 42 seconds on a laptop with 8 GB of memory. Internet is needed only to fetch ground and laser data for a new scene.
2. Inside existing mapping work. The outputs are standard files, so NDMA and state disaster authorities can load them into the software they already use.
3. Through a phone page for field staff. On the same network, a phone records GPS points and known heights, which return to the app as check points.
4. Through a web demo anyone can open. The repository runs as a hosted demo on Replit with a processed scene, so a visitor can explore the tools in a browser without installing anything. A check every five minutes keeps it awake.
5. Through the Mamba model on a central server. A faster model lets one graphics card process many districts, with the results sent to offices as files.

Each rollout would start with a test on Indian imagery, then a pilot with one office through a monsoon, staff training with the user guide, and feedback through the app's own checking page. Because the checking page reports errors in the same way for every scene, feedback from a pilot can be compared directly with the tests in this report.

9. Possible Constraints and Barriers to Implementation; Design Issues

Table 5 lists the limits we know of. The largest is accuracy on tall objects. Design choices follow from these limits: heavy data is stored per scene so the tool works offline, every disaster output is labelled a scenario, and a faster model must match the current one on the same tests before it replaces it. Population figures come from building floor area, not from personal data.

The data we check against has limits of its own. Most laser tracks were recorded years before or after the photograph, so a new building can look like an error. India has only two test sites for each kind of landscape, which makes the confidence ranges wide. And the laser's own labels can be wrong in dense towns, which is why the roof-point rule exists.

Table 5: Known limits and what DepthWizard does about each today.

Limit	Effect	What we do
Objects over 20 m are 15 to 18 m too low	Tall refuges look shorter	Warning on every result
Trees at 18 to 60% of true height	Forest heights unreliable	Tree-height map or laser correction
Ground under forest about 20 m off	Floods in forest unreliable	Flagged in the app
Mamba model less accurate so far	7.20 against 3.90 m on aerial tiles	App keeps RS3DAda
Mamba model needs an NVIDIA graphics card	Not in the laptop version	Server use only
Only two Indian test sites per landscape	Wide confidence ranges	Ranges always reported
India's Cartosat images need a login	Not yet tested	Request to NRSC drafted
Some data is for non-commercial use only	Commercial use restricted	Research and government pilots

10. Expertise Available with Each Student to Contribute to the Development of the Intervention

The project drew on machine learning, satellite data and web development, and just as much on research, writing, testing and presentation. Table 6 lists what each member brought to the team.

Table 6: Expertise of each student.

Student	Expertise	Student	Expertise
Himanshu Patel	Full-stack web development and hosting, machine learning, satellite and map data	Saloni Mahajan	Writing, editing and figure layout
Shlok Vaidya	Model training, testing and benchmarks	Peruru Smitha Reddy	Presentation design and communication
Agampreet Singh	Research on floods and landslides in India	Akshita	Documentation and user guides
Saaransh Garg	Software testing and user feedback	Aryan Goyal	Data handling and literature review
Takshita Nagrale	Technical writing and editing	Arahana Garg	Human geography and disaster management

11. Expected Outcomes

The main outcome is an offline tool that gives a checked height map from one image. We tested it on places the model never saw in training: 150 aerial tiles of three US cities with aircraft laser heights from the GAMUS set [19], 300 satellite tiles each of Atlanta and Buenos Aires from Overhead Geopose [20], and eight Indian sites checked against NASA lasers. Each number is an average with a 95% confidence range in brackets, set against a simple guess at the same points.

11.1 Height accuracy

Table 7: Average height error in metres with RS3DAda (lower is better).

Test	Size	DepthWizard	Simple guess
Aerial tiles, every point	150 tiles	3.90 [3.24, 4.66]	5.65 (flat ground)
Aerial tiles, buildings	1,450	3.57 [3.08, 4.06]	10.54 (flat ground)
Atlanta, satellite	300 tiles	5.99 [5.57, 6.43]	5.67 (published)
Buenos Aires, satellite	300 tiles	1.94 [1.86, 2.03]	2.01 (published)
India, against ICESat-2	8 sites	4.85 [3.04, 6.69]	8.25 (height 0)
India, against GEDI	7 sites	9.93 [4.86, 15.42]	16.89 (height 0)

Table 7 shows that for buildings, DepthWizard is about three times closer to the truth than guessing flat ground. At Indian sites it is almost twice as close as guessing zero. On satellite images it matches the results the original authors published. Figure 5 shows where the gain comes from: buildings and trees. On roads and bare ground the true height is near zero, so the flat guess is already right. The same pattern shows by landscape on the aerial tiles. In towns the error is 2.90 m against 5.66 m for the flat guess, and in forest 6.67 m against 8.70 m. On open land the flat guess wins, 0.55 m against our 0.96 m. Reading each tile four times instead of once lowers the error by 0.064 m on average and helps on 114 of the 150 tiles, so we kept it.

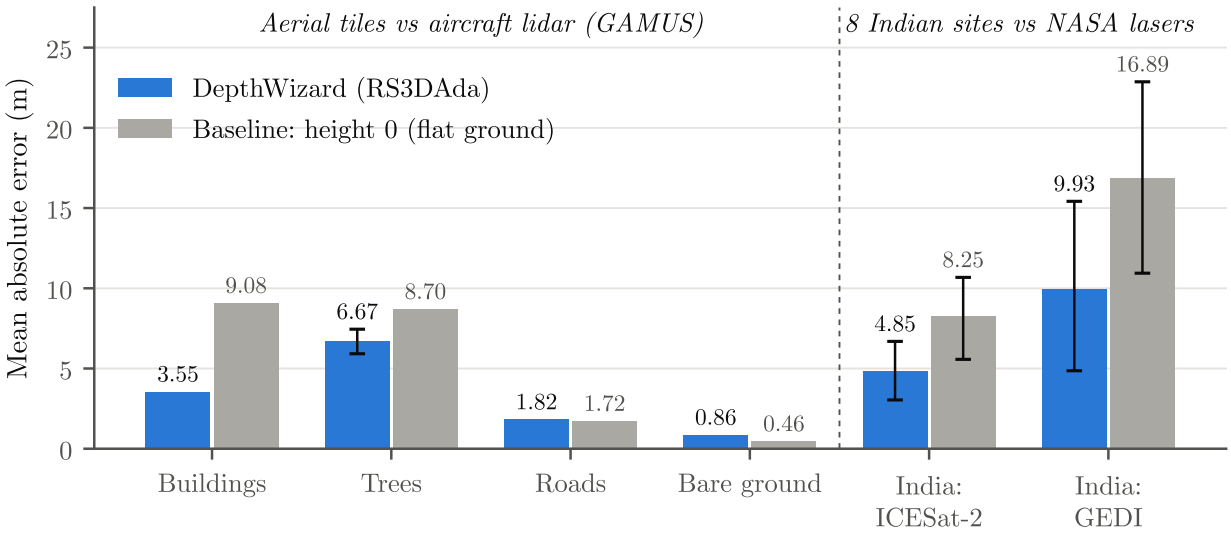


Figure 5: Average height error by type of surface. Left: aerial tiles against aircraft lasers. Right: Indian sites against NASA lasers, with 95% confidence ranges.

11.2 The Mamba model and other models

We ran four models on the same 750 tiles: RS3DAda, our Mamba student, and two versions of Depth Anything V2 [6]. Depth Anything does not give heights in metres, so we also score every

model after fitting its output to the truth with the best scale and offset for each tile; Table 8 calls this score "fitted". It shows how well a model finds where the tall things are.

Table 8: Height models on the same 750 tiles: average error in metres, and cost on one graphics card.

	As predicted			Fitted		
Model	Aerial	Atlanta	B. Aires	Aerial	Atlanta	B. Aires
Flat ground	5.65	8.98	3.86			
RS3DAda	**3.90**	**5.91**	**1.86**	**2.95**	**4.94**	**1.79**
Mamba student (ours)	7.20	8.70	3.39	3.97	7.10	2.54
Depth Anything V2, relative				3.25	5.64	2.18
Depth Anything V2, metric				3.49	6.21	2.40

	Size	Memory	Speed
RS3DAda	366.5 M	705 MiB	105 tiles per minute
Mamba student (ours)	40.7 M	78 MiB	546 tiles per minute
Depth Anything V2, relative	335.3 M	1,532 MiB	118 tiles per minute
Depth Anything V2, metric	335.3 M	650 MiB	134 tiles per minute

The Mamba student is the cheapest model by a wide margin. It is nine times smaller than RS3DAda, uses nine times less memory and runs 5.2 times faster. Its first version is less accurate: 7.20 m against 3.90 m on the aerial tiles, where it is worse than the flat guess, and 8.70 m and 3.39 m against 5.91 m and 1.86 m on the satellite tiles, where it is slightly better than the flat guess. The app therefore keeps RS3DAda for now. RS3DAda is also more accurate than Depth Anything V2 on all three sets. The gaps are small, 0.30 m, 0.70 m and 0.39 m for the relative version after fitting, but all three are outside the confidence ranges.

The student's main weakness is in finding where the tall things are. Even after fitting, its heights follow the true ones much less closely than RS3DAda's, with a correlation of 0.29 against 0.61. On the aerial tiles it also makes low ground about 3 m too high. Both problems should shrink with longer training and with true heights in the training goal, which is the first item in Section 12. Figure 6 shows typical tiles.

11.3 Tall objects and error ranges

Low objects are about right, but tall ones come out far too low, as Figure 7a shows. Above 20 m, heights are 16.26 m too low on aerial tiles and 18.31 m too low in India. The two satellite cities show the same thing, at 16.80 m and 15.25 m too low. This is the main weakness of the published model, and our Mamba student inherits it. The 90% error range held for 86.6%, 83.3% and 97.0% of building blocks in three cities it was not set on, with a width of about 6 m either side. On 96 Indian roofs it held for 71%. Figure 7b shows that it fails for buildings over 20 m, and the app warns users about this.

11.4 Ground level and floods

Removing rooftops and trees from the free ground map halves its error in Indian towns, from 5.28 m to 2.36 m, as Figure 8a shows. In hills FABDEM is better, and under forest every map is about 20 m off. Leveling with lasers removes the map's average offset, from 2.58 m too high to 0.27 m too low. The full surface, ground plus objects, is closer to GEDI's tree-top and roof-top heights than the free 30 m surface map is, at 5.97 m against 8.39 m. Against every laser point, ground and tops together, it is not: 9.14 m against 8.06 m. What the photograph adds is detail at half a metre, which a 30 m map cannot have.

That offset matters most for floods. We flooded the pre-flood heights of four areas of Porto Alegre,

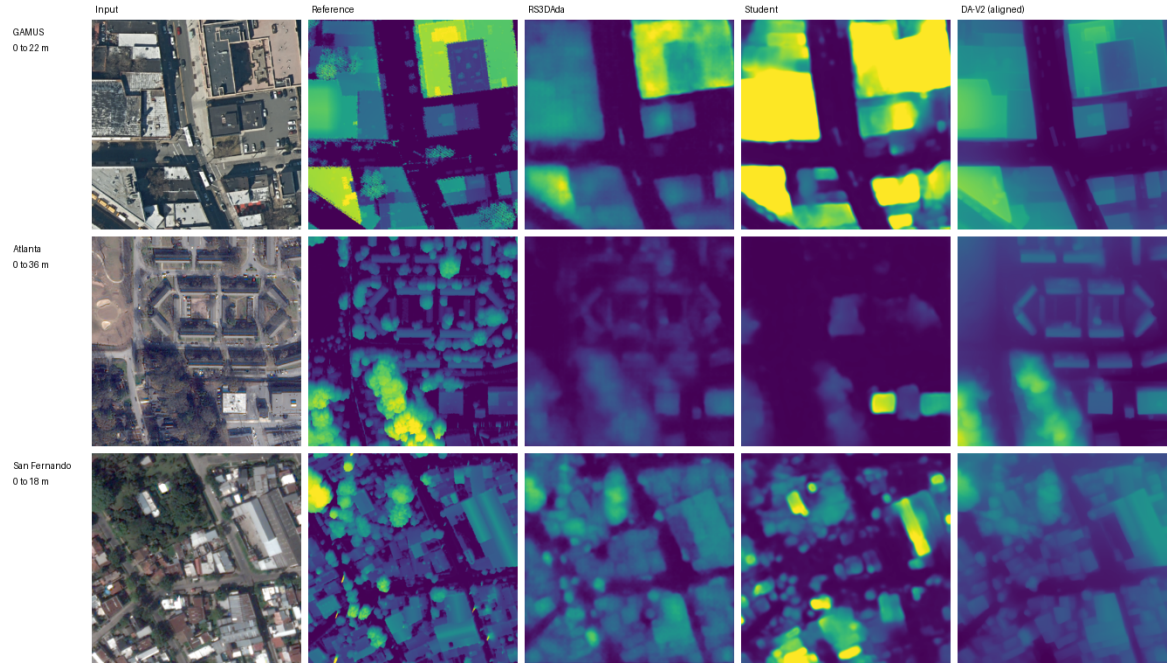


Figure 6: Typical tiles: input, laser truth, RS3DAda, our Mamba student and Depth Anything V2 (fitted). Rows: aerial set, Atlanta, Buenos Aires set. Dark is low and yellow is tall. The student finds buildings but makes them too tall on the aerial tile and blurs small objects.

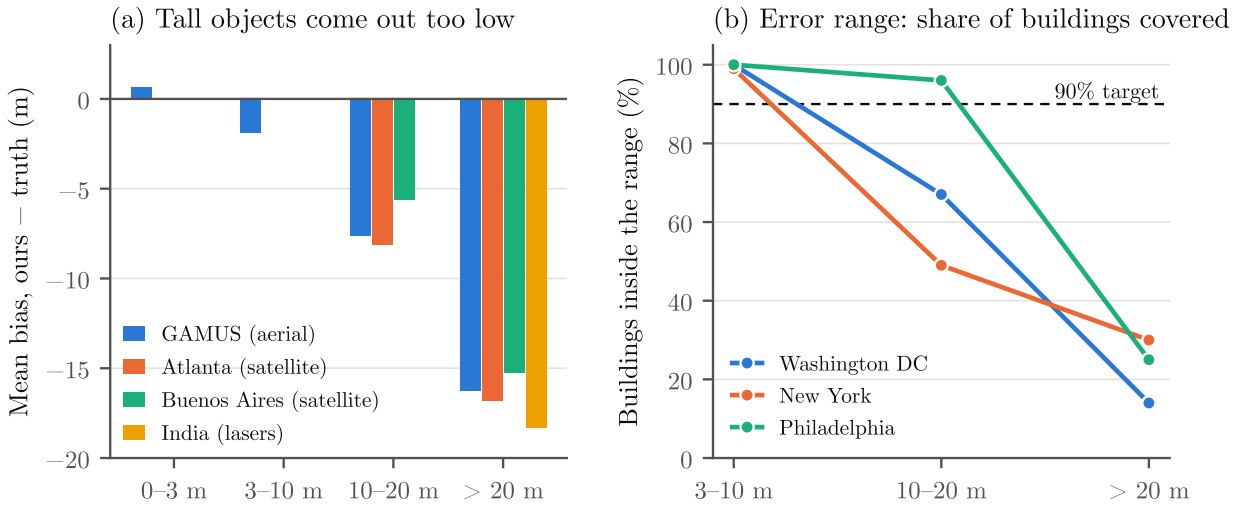


Figure 7: (a) Average error by true object height. (b) Share of buildings whose true height falls inside the 90% range, by height.

Brazil, and compared them with the mapped flood of 6 May 2024. The match, measured by CSI, was 0.20 with the free ground map, 0.30 with our cleaned map and 0.79 after laser levelling. Figure 8b plots these scores. FABDEM scored 0.32. Over all seven areas we tried, without laser levelling, the scores were 0.18 for the free map, 0.47 for ours and 0.48 for FABDEM. The areas are small, so this is a test of the method and not a forecast.

11.5 A real Indian disaster: the Teesta flood, Sikkim, 4 October 2023

On 4 October 2023 the South Lhonak glacial lake burst and sent a flood down the Teesta valley. Using free public data only, we compared Sentinel-2 [22] satellite images from November 2022 and November 2023, the first cloud-free pair after the flood. Ground near the river that lost its vegetation or turned into water counts as damaged. We found 1.281 km^2 of damaged ground along

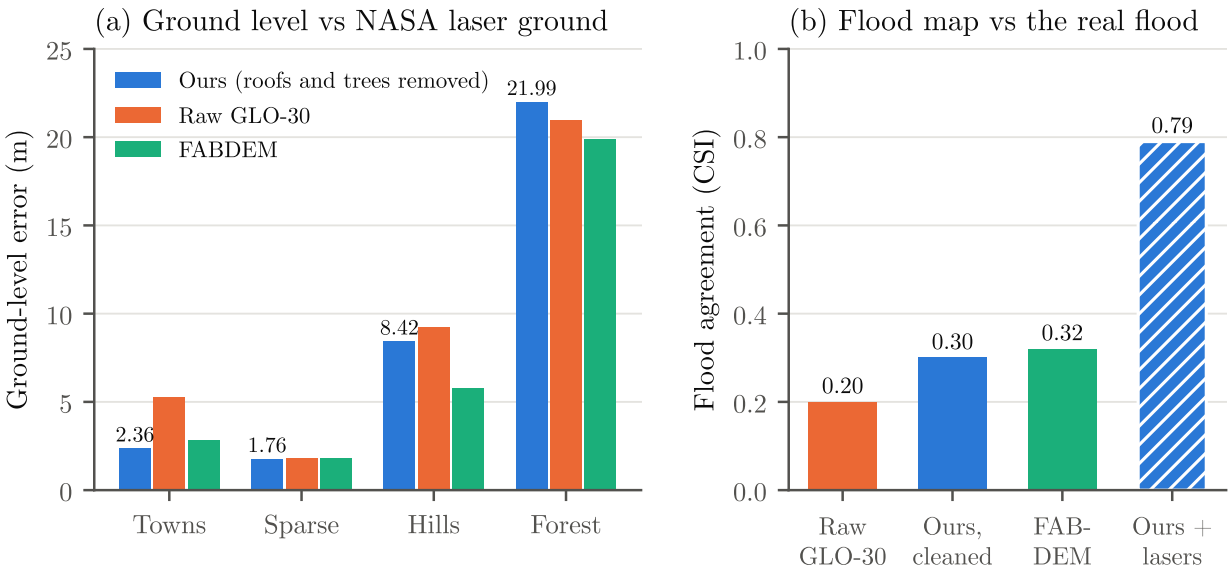


Figure 8: (a) Ground-level error against NASA laser ground at the Indian sites, by landscape. (b) Match between predicted and real flood maps in Porto Alegre with four ground maps. Striped: our map levelled with lasers.

12 km of valley, mapped in Figure 9. The lake itself shrank by only 5.7%, from 1.358 to 1.28 km^2, but its water became much muddier.

We then ran DepthWizard on a pre-flood photograph of the valley [21] and loaded the damage map into the Observed-flood tool, shown in Figure 3f. The best-fitting flood scenario matches the damage map with a CSI of 0.49. It covers 86% of the damaged ground, but 47% of the ground it floods was not damaged. A single water level is a rough model for a fast flood in a steep valley, which explains much of the over-flooding. The damaged area reached 52 buildings, home to about 673 people by the app's estimate. Two cautions apply. The first clear images came a month after the flood, so the map shows what the flood left behind and not its peak. And each satellite pixel covers 10 m, which suits areas but not single houses.

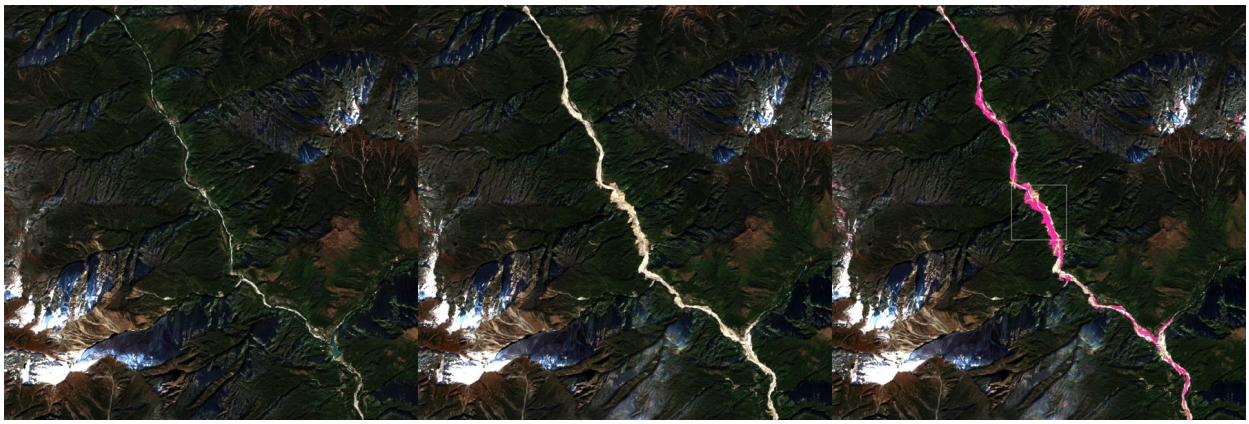


Figure 9: The Teesta valley before (left) and after (middle) the flood, and the damaged ground in magenta (right). White box: the area loaded into DepthWizard.

11.6 What we tried that did not help

Some ideas were tested on the same data and dropped. A correction curve for heights helped in two cities and hurt in the third, where the error rose from 1.86 m to 2.87 m. Resizing the input image made no clear difference. Using GEDI points to make the laser levelling denser was slightly

worse than using ICESat-2 alone. An error range for every pixel, in place of every building, held between 81.4% and 96.1% of the time across cities and missed its 90% target, so the app shows ranges for buildings only.

12. Suggested Plan of Action for Utilization of the Expected Outcome from the Work

The plan runs in four phases over about two years. Each phase starts only when the one before it has met its check, so effort goes where the evidence says it is needed.

Table 9: Plan of action, with the check that ends each phase.

Phase and time	What we will do	Check before moving on
1. Fix the model 0 to 6 months	Train the Mamba model for longer, add true heights from the synthetic cities, and adapt it to real images. Correct tall buildings from shadow length and sun angle. Measure the tree-height corrections on Indian forest sites.	Mamba error within 0.5 m of RS3DAda on the 750 test tiles; error above 20 m cut by half
2. Test on Indian data 3 to 9 months	Get Cartosat images and CartoDEM from NRSC and repeat every test. Add at least ten sites for each landscape, with ground points from a state survey department.	Results hold on Indian imagery; confidence ranges narrow enough to compare landscapes
3. Pilot with a state one monsoon	Work with one State Disaster Management Authority, for example Assam or Himachal Pradesh. Before the rains, give flood-reach and landslide maps for two or three districts. After each flood, compare them with the mapped extent and with field reports.	District officers use the maps in their monsoon plans; flood match (CSI) of 0.6 or more on real events
4. Hand over and scale from year 2	Give the code, test register and user guide to NRSC and ISRO so the model can run on their servers and feed Bhuvan. Train district staff with short hands-on sessions, and offer the tool to geography and planning courses.	A state or national agency runs it without our help

Alongside disaster work, the same height maps can help with town planning, rooftop solar studies and checks on unauthorised construction. These need the same thing, a cheap height map with a known error, so they can be added at any phase without new research.

13. Conclusion

One ordinary satellite photograph, combined with free ground data and NASA's lasers, gives a 3D height map that disaster teams can use offline and check for themselves. DepthWizard beats simple guesses on every height test with the published model, halves the ground-level error in Indian towns, and maps a real Indian flood from public data. Our Mamba model is nine times smaller and five times faster than the published one, and making it as accurate is the next job. The tool does not replace a ground survey yet: very tall buildings come out too low and forests remain hard. Every result carries its error, so users can see where to trust it.

Data and code

The code, the register of every number, a video walkthrough and the steps to run the web demo are in the project repository: https://github.com/hima1323/terravista-depthwizard. All test data are public.

14. Contribution of Each Student of the Group to Complete the Assignment

Table 10: Contribution of each student to the project and this report.

Student	Contribution
Himanshu Patel	Led the project. Built the whole website and app, front end, back end and hosting; the ground-level correction; the Mamba model code; the eighteen tools; the Teesta case study; and wrote this report
Shlok Vaidya	Built checking in the app and survey-point correction, the installer, and the helipad and route tools; trained the Mamba model and ran the model comparison
Agampreet Singh	Gathered reports on past floods and landslides in India for the problem definition
Saaransh Garg	Tested the demo on the sample scene and reported interface problems
Takshita Nagrale	Edited the report and checked it against the HS202 format
Saloni Mahajan	Helped select and arrange the figures and screenshots
Peruru Smitha Reddy	Designed the presentation slides
Akshita	Reviewed the README and the demo instructions
Aryan Goyal	Checked the numbers in the result tables against the source files
Arahana Garg	Summarised disaster-management guidance used in the background section

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